

FOOTWEAR FIELDS: Learning a Functional 3D Space for Topology-Varying Footwear

Internal Project Report and Research Plan

Visual Intelligence Lab (VILab), Drexel University

August 2026

Working slogan: Shared Anatomy, Without Shared Topology.

1 Topic

We are building the first anatomy-grounded category-level 3D model of footwear. The central idea is that shoes of radically different shapes and topologies (sneakers, Crocs with holes, sandals made of straps, slippers, high heels, boots) do not share a common mesh structure, but they all serve the same human foot anatomy. We therefore represent every shoe not in ordinary object-centric coordinates, but in a coordinate system defined by a parametric 3D foot model (SUPR-Foot). In this shared functional space we learn a single generative model of footwear geometry and appearance, a notion of correspondence between shoes that is allowed to say “this part does not exist on that shoe,” and a principled way to complete the parts of a shoe that retail photography never shows (the sole, the top opening, the interior).

The target venue is CVPR. The deliverables are: a method (the representation and model), a dataset (4,000+ real shoes organized in the shared functional space, plus a smaller fully scanned benchmark set), and a benchmark suite of five tasks.

The scientific question, stated once and referred to throughout:

Can shared human anatomy provide a common 3D space for wearable objects that share function but do not share topology?

Footwear is our testbed for this question. The introduction of the paper will note that the same principle extends to gloves and hands, eyewear and faces, helmets and heads. We will not attempt those in this paper; they are the vision statement, not the scope.

2 Motivation

2.1 The data and its blind zones

We have collected 36-view ring captures of more than 4,000 shoe instances from StockX product photography. All 36 views come from a single camera ring at one fixed elevation. This has an important consequence: three regions of every shoe are *never* observed, no matter how many views we use: the outsole, the top-down view of the opening, and the interior cavity. This is different from ordinary sparse-view reconstruction, where missing regions are an accident of view count. Here the

The parts of a shoe that the camera never sees are also the parts that are most closely related to the shape of the human foot. The inside of the shoe has to leave enough space for a foot. The sole sits directly underneath the foot.

The opening has to be large enough for the foot and ankle to enter the shoe. Because of this, when we need to estimate parts of the shoe that are missing from the images, a human-foot model gives us a more meaningful guide than a generic image or 3D generative model. This makes the foot model a central part of our method, rather than simply an extra constraint.

missing set is systematic, identical across all 4,000 instances, and analytically characterizable from the ring geometry. We call this setting *ring-observed reconstruction*. Adding more views on the same ring never fixes it.

2.2 Why anatomy is the right prior

The regions the camera never sees are exactly the regions the foot determines most strongly. The interior cavity is shaped around a last, which is itself an idealization of a foot. The outsole is the plantar interface. The opening admits an ankle. So the correct prior for completing what the camera never observes is not a generic image or 3D diffusion prior; it is an anatomical one. This observation upgrades the foot model from a gimmick to the central argument of the paper.

2.3 Why topology forces a new representation

A naive implementation would represent each shoe as a displacement shell over the foot surface. This fails immediately on our data: sandals, Crocs, slippers, and laced shoes have material distributions that cannot be described as a single closed shell around the foot. A ray leaving the foot may pass through cavity, upper, air, lace, and air again. The representation must allow material to exist or not exist at any anatomical location. Absence of material is itself meaningful: it is what distinguishes a sandal from a sneaker.

2.4 Why this is high impact

Four user communities, one sentence each. E-commerce and virtual try-on: industry fit systems (Volumental, Neatsy, Amazon's Shoefitr acquisition) rely on proprietary last data; our model recovers an anatomical envelope from public exterior imagery, sidestepping the manufacturer data wall. Embodied AI and digital humans: current 3D asset libraries contain shoes as hollow-less shells; ours are the first assets with anatomically consistent interiors, i.e., shoes that can actually be worn by a digital body. Forensics: shoeprint analysis needs outsole geometry, which our model predicts from views that never show the sole. Biomechanics: the shoe-foot interface is a long-standing missing element in gait simulation, and this project connects directly to the lab's broader program on functional human modeling. How a person walks is called gait simulation.

I think there was paper for Forensics in the other document. I need to check that

3 Existing Methods and Where We Stand

We ran two systematic literature surveys (positioning and tooling). The condensed picture, organized as four blocks plus supporting areas:

3.1 Parametric foot models (tools we build on)

SUPR-Foot (Osman et al., ECCV 2022) provides an articulated parametric foot with pose and shape spaces and learned ground-contact deformation; it was trained with a scanner that images the sole through glass. FIND (Boyne et al., BMVC 2022) is an implicit foot model with the Foot3D scan dataset; FOUND (WACV 2024) and FOCUS (3DV 2025) reconstruct feet from images. All of this work treats the foot as the reconstruction target. Our reversal: the foot is not the target, it is the coordinate system and prior for another object. No prior work uses a foot model this way.

I would need to understand more about how we can use the foot's coordinate system to model a shoe and also use it as a prior.

It would give me some ideas to check how SMPL use used as a conditioning model for garmets.

3.2 Body-anchored garment representations (closest methodological family)

SMPLicit (CVPR 2021) represents varying-topology garments as an unsigned distance field conditioned on SMPL. DrapeNet (CVPR 2023) generates and drapes garments of arbitrary topology from latent codes. ISP (NeurIPS 2023) represents garments as 2D sewing panels with a learned 2D-to-3D mapping, which is the strongest precedent for our UV-domain coverage idea. The 2024–2026 wave (DressCode, SewFormer, GarmentCode, ReWeaver, PatternGSL) moves toward structured, simulation-ready garment representations. The critical limitation of the entire family: every method models a thin cloth shell. None models a thick, volumetric, rigid-and-soft composite object, none models an interior cavity, and none treats absence of material as a first-class variable. Our answer to the inevitable reviewer question “why isn’t this SMPLicit for shoes” must be complete: garment methods represent a surface worn around a body; we model a topology-varying material volume organized by anatomical function, where absence of material is meaningful and the wearer anatomy provides partial correspondence across objects that do not share surface support. A useful technical point: garments were forced into unsigned distance fields because cloth is an open zero-thickness surface with no defined inside. Shoes are solid volumes, so a material SDF with a well-defined sign works. The volumetric nature of footwear rescues the signed representation.

3.3 Shoe-specific work (nearly empty field)

The closest work is a 2025 arXiv paper doing per-instance 3D Gaussian Splatting of shoes for mobile AR, with a small segmentation dataset. ShoeRinsics (WACV 2023) does intrinsic decomposition of shoe treads for forensics. ShoeModel (ECCV 2024) is 2D diffusion generation of shoe-wearing advertising imagery. There is no academic parametric or morphable 3D shoe model, and no large real-capture 3D shoe dataset. We will phrase this carefully (“to our knowledge, existing footwear-specific methods focus on per-instance reconstruction, visualization, try-on, or outsole analysis rather than a shared category-level 3D representation”) because it is a negative search result and arXiv moves fast; we will re-run the scoop check two to three weeks before the deadline.

3.4 Generic 3D generation and completion (the real competition)

TRELLIS (CVPR 2025) and its successors, GS-LRM (ECCV 2024), and occlusion-aware models like Amodal3R (ICCV 2025) are the true scooping risk: a reviewer can always ask “why not fine-tune a 3D foundation model on shoes?” Our defense is threefold. First, these models condition on visibility masks or generic category priors, not anatomy, and enforce no interior structure. Second, they fill *random* occlusions, whereas our missing regions are systematic; nobody has an evaluation protocol for never-observed regions, and we will introduce one. Third, we will run the fine-tuned foundation model ourselves as a baseline row in the main table, so the comparison is empirical, not rhetorical.

3.5 Supporting areas

Body-part-anchored objects beyond garments exist and support our generalization claim: MEGANE (CVPR 2023) anchors a morphable eyeglass model to a face avatar; HOLD and G-HOP (CVPR 2024) anchor hand-held objects to MANO. There is no helmet-to-head precedent, so the intro cites eyewear and hand-object work only. On correspondence, partial functional maps (Rodolà et al.) already formalized matching where correspondences may not exist; our novelty is the *source* of partiality (functional coverage differences over shared anatomy, not shape dissimilarity), and we must cite this line explicitly to preempt the “unaware of partial matching” review. Large multi-

view datasets (ABO, MVImgNet/2.0, OmniObject3D, Objaverse-XL) provide scale but no category structure, no correspondence, no body anchoring; ABO (CC BY 4.0) is our licensing precedent. On reconstruction tooling, our second survey selected: parametric ring-fit pose solving (exploiting known uniform azimuth, initialized by masked VGGT or MAST3R-SfM; classic SfM is documented to fail on turntable captures), PGSR or 2DGS as the surface backbone, Splatfacto-W-style per-image appearance embeddings to absorb the rotating studio lighting, UNISURF/NeuS-style volume rendering losses for field supervision, DCUDF-family meshing as export-only (UDF zero-level-set extraction is still fragile and must not be a load-bearing dependency), and TRELLIS/Amodal3R fine-tunes as baselines (feasible on a single 24GB GPU, roughly one GPU-day per fine-tune).

4 Proposed Approach

4.1 Overview: one function, three capabilities

One equation is enough for the whole paper. A shoe is a function

$$D(\Phi_F(x), z),$$

where x is a 3D point, Φ_F maps it to anatomical coordinates relative to a fitted SUPR-Foot instance F , z is a latent code identifying the shoe, and D is a signed distance to the material boundary: negative inside shoe material, positive in air. Everything else is a consequence of this choice.

Three consequences, in plain language. First, topology is free: holes, straps, open heels, thick midsoles, and disconnected parts are all just patterns of where material exists, and an SDF never requires two shoes to share a mesh. Second, correspondence is free: the query “lateral forefoot, 15mm out” names the same functional location on a Jordan, a Croc, and a sandal, and the model may answer “material” or “no material.” Third, completion is constrained: once thousands of shoes live in one aligned space, the distribution of hidden regions given visible regions becomes learnable, with the anatomy prior carried by the coordinates and the style prior carried by the latent.

The rest of this section specifies each component at the level of detail needed to start implementing: how the foot is fitted (4.2), how the coordinate chart is constructed and where it can break (4.3), how the material field is structured (4.4), appearance (4.5), the latent space and category prior (4.6), supervision in both the mesh-supervised and image-supervised regimes (4.7), how correspondence is operationalized (4.8), the full image pipeline (4.9), and design risks with fallbacks (4.10).

4.2 Fitting the foot: from a shoe to its anatomical frame

Every shoe, whether it arrives as a CAD mesh (Stage 1) or as reconstructed surface evidence from images (Stage 3), must first receive a fitted SUPR-Foot instance $F = (\beta, \theta, s, T)$: shape parameters, pose, scale, and a rigid transform. The fit defines the anatomical frame, so its conventions must be deterministic; any freedom here misaligns the shared coordinate domain across shoes and silently corrupts everything downstream.

Normalization first. Detect the ground plane from the outsole contact region, orient gravity, find the heel-toe axis from the footprint’s principal direction, and normalize scale by insole length. All shoes are mirrored to a canonical side (left), exploiting bilateral symmetry.

Pose by convention, not by optimization. Heel height and foot plantarflexion are entangled: the same shoe can be explained by different (pose, geometry) pairs. We remove this degree of freedom with a fixed rule: measured heel drop (height difference between heel and forefoot support

By representing shoes as foot-aligned material fields, diverse geometries are handled uniformly. Corresponding locations are well-defined across styles, and completion of unseen regions is learned under anatomical and style constraints.



using PCA might cause some problems with boots. Need to check what work around I did for alignment

Before fitting the supr-foot, normalize every shoe mesh. Same ground plane, same up direction, align heel-toe consistently, scale by insole length. Then fit the reference foot inside.

planes) maps through a deterministic table to a plantarflexion angle. Toe splay and other pose parameters are frozen at a neutral standing pose. The rule is part of the definition of Φ_F , is stated in the paper, and is stress-tested on the high heels in the CAD set during week one. For boots, the anatomical domain is extended upward with a leg segment taken from the full SUPR body model, so the shaft has valid coordinates.

Fitting objectives. With pose fixed by convention, the remaining fit (shape β , scale, transform) minimizes three terms: a containment term penalizing foot surface points that fall inside shoe material (the foot must live in the cavity’s free space); a plantar contact term pulling the sole of the foot toward the insole surface; and a clearance regularizer keeping a small, anatomically plausible gap over the dorsal surface. A shape prior keeps β near the population mean unless the shoe demands otherwise. For open shoes (sandals, slippers) the containment term applies only where material exists, which is exactly what the coverage function (4.4) provides; in Stage 1 we bootstrap this from CAD part structure.

What the fit is not. The fitted foot is a reference anatomy, not a claim about the true last or about any particular wearer. This distinction is stated in the paper and repeated in the limitations.

4.3 The anatomical coordinate chart Φ_F

Φ_F assigns each 3D point x a triple (u, v, r) : a location on the foot surface (which functional region, medial or lateral, plantar or dorsal) and a scalar offset describing how far outside the foot the point lies. The naive construction, nearest point on the foot plus Euclidean distance, is correct near the foot and breaks exactly where shoes need it most. In concave regions (the arch, the ankle notch, between the toes) nearest-point rays cross, so multiple (u, v, r) triples describe the same x and the field value at x is ill-defined. Far from the foot the map degenerates: a high heel block occupies a large volume whose nearest-point projections all collapse onto a small patch of heel skin, so the chart compresses catastrophically. The valid region of the naive chart ends roughly at the medial axis of the foot’s exterior, and real shoes (heels, platforms, thick midsoles) have plenty of material beyond it.

Three candidate designs will be implemented and compared in Stage 1; choosing among them is a method decision, and the analysis itself can become part of the contribution.

1. **Generalized nearest-point coordinates with union semantics.** Keep the naive chart but make multiplicity explicit: a point may carry several candidate coordinates, the field is evaluated at each, and material is the union (smooth minimum of the SDF values). Simple, fully differentiable, no precomputation; the cost is that the decoder must be robust to duplicated queries and the chart remains compressed at the heel block.
2. **Harmonic coordinate field.** Solve a Laplace problem in the volume around the foot, with boundary conditions given by the foot surface parameterization on the inner boundary and a far enclosing surface on the outer boundary. The solution propagates (u, v) smoothly into the volume and yields a level-set coordinate playing the role of r . Harmonic interpolation gives a smooth, crossing-free chart by construction, at the price of a per-foot-shape numerical solve (amortizable, since the fitted feet vary in a low-dimensional shape space).
3. **Learned diffeomorphic mapping.** An invertible network, conditioned on the SUPR-Foot parameters, maps world space to a canonical foot-centered space; train with cycle-consistency and distortion penalties. Most flexible and handles the heel block gracefully; heaviest to train and hardest to interpret.

Selection criteria, measured on the CAD set: fraction of shoe-occupied volume where the chart is non-injective; metric distortion statistics inside material regions, with the heel block reported separately; and downstream Stage-1 reconstruction and generalization scores under each chart with the decoder held fixed. The default expectation is that option 1 suffices for sneakers and sandals and option 2 or 3 is needed once heels and platforms enter; the paper reports the comparison either way.

4.4 The material field: a shell-plus-residual decomposition

A completely free SDF decoder would technically suffice, but it discards the strongest prior we have: the cavity is foot-shaped and the shoe body is, to first order, a thickened partial shell around the foot. We therefore structure D as a decomposition, and ablate the structure explicitly.

Base shell. Three maps on the 2D foot domain define it: a coverage function $C(u, v) \in [0, 1]$ answering “does this shoe have material over this anatomical region”; an inner offset $o_{\text{in}}(u, v)$, the small clearance between foot and cavity wall, tightly regularized because it encodes the anatomical envelope; and an outer offset $o_{\text{out}}(u, v)$, the material thickness, which is large under the plantar region (midsole and outsole) and small on the upper. Where C is high, the shell contributes material between the two offset surfaces; where C is low, it contributes nothing. The decisive property: *all topology variation concentrates into the 2D coverage map*. Crocs holes are holes in C ; a sandal is C collapsed to straps; a slipper is C nonzero only over the forefoot; a boot extends C up the leg segment. Coverage, offsets, and everything downstream live on a regular 2D domain where convolutional and diffusion architectures are comfortable, which is the same reason ISP’s sewing-panel parameterization works for garments; the shoe-specific twist is that our 2D domain is anatomical rather than a designer’s flat pattern, and our shell has volume.

Residual components. A small number of sparse additive elements handle what the shell cannot: straps that leave the foot surface’s offset family (a heel strap bridging through air), buckles, decorations, and the heel block itself if the chosen chart handles it poorly. Each residual is a localized SDF primitive in anatomical coordinates; the full field is the smooth union of shell and residuals. Sparsity is enforced so the shell explains as much as possible and residuals stay interpretable.

Sign, and why shoes are easier than garments here. Garment work was forced into unsigned fields because cloth is an open, zero-thickness surface with no defined interior. Shoe material is a solid volume; inside and outside are well defined everywhere, so we keep a signed field and standard watertight extraction. This point goes in the paper: the volumetric nature of footwear rescues the signed representation.

Planned ablation. Free SDF (no structure) versus shell-only versus shell-plus-residual, all with identical decoders and latents, scored on reconstruction, leave-one-topology-out generalization, and interpolation plausibility. This table justifies the decomposition rather than asserting it.

4.5 Appearance

Appearance lives in the same coordinates: an albedo field over the material boundary, parameterized by (u, v) for the shell surfaces (giving every shoe a texture in a shared anatomical UV space, which is what makes cross-shoe texture transfer well-posed) and by local coordinates for residual elements. View-dependent effects in the training images are absorbed by per-image appearance embeddings during fitting (4.7) and are not baked into the asset. Relightable material decomposition (albedo, roughness, normals) is an upgrade attempted only after the core is stable, and never a headline

claim.

4.6 Latent structure and the category prior

The per-shoe code factorizes as $z = (z_{\text{struct}}, z_{\text{geom}}, z_{\text{app}})$: structure drives coverage and residual existence (what kind of shoe this is, topologically), geometry drives offsets and residual shapes, appearance drives texture. The factorization is encouraged, not hard-enforced: swapping z_{app} between shoes should re-texture without changing shape, and interpolating z_{struct} should morph a sneaker toward a sandal by dissolving coverage, which is exactly the Figure 4 demonstration.

Training proceeds in the auto-decoder style of DeepSDF: each training shoe owns a code optimized jointly with the shared decoder. After the codes exist, a generative prior (a latent diffusion model, or a simpler Gaussian mixture if the population is small) is fitted over them. This prior is what turns the representation into a category model: sampling gives new shoes, latent walks give topology-crossing interpolation, and completion becomes posterior inference, recovering the distribution of codes consistent with partial evidence and decoding hidden regions from them. The anatomy prior constrains where things can be; the style prior, carried entirely by z , determines what they look like. The foot tells you nothing about an AJ1 tread pattern, and the formulation never pretends it does.

4.7 Supervision

Mesh-supervised regime (Stage 1, CAD and scans). Sample query points densely near the ground-truth surface and uniformly in the volume; obtain ground-truth sign for imperfect CAD meshes with fast winding numbers rather than mesh repair; regress D with a clamped SDF loss plus an eikonal regularizer. Where CAD part structure exists (sole, upper, strap as separate components), bootstrap supervision for C and for residual assignment from part names; this is weak supervision and is treated as such. Functional landmarks (toe tip, heel center, malleoli projections) add a light consistency loss tying the chart across shoes.

Image-supervised regime (Stage 3, ring captures). The field is rendered differentially by converting the SDF to occupancy and volume-rendering it in the UNISURF/NeuS style, with a sharpness schedule annealing from soft to hard. Losses: a silhouette term against segmentation masks; a photometric term with per-image appearance embeddings absorbing the rotating studio illumination; and, critically, analytic visibility weighting: the ring geometry tells us in closed form which surface normals are observable, data terms apply only there, and unobserved regions (sole, top, interior) are governed purely by the latent prior. This is the mechanism that makes “completion is an application of the representation” literal rather than rhetorical: nothing special happens at the blind zones, the prior simply keeps holding where data terms fall silent.

4.8 Existence-aware functional correspondence, operationalized

Coverage gives existence directly: $E(u, v) = [C(u, v) > \tau]$. A point on shoe A maps through the chart to (u, v) , and its counterpart on shoe B is the material at the same (u, v) if $E_B(u, v) = 1$, else $\emptyset$. Region-level correspondence transfers functional-zone masks (plantar support, toe enclosure, heel enclosure, dorsal coverage, heel retention, ankle enclosure) through the shared domain. Evaluation uses the annotated benchmark from Stage 2: region transfer accuracy where the counterpart exists, and existence-prediction F1 where the question is whether it exists at all. This is the formal object behind the Figure 2 demonstration, and it is what separates the paper from partial-matching

work driven by shape similarity: partiality here is a statement about functional coverage of shared anatomy.

4.9 Pipeline from raw StockX images to assets

Stage by stage: (1) segmentation and left-right mirroring; (2) camera recovery by parametric ring fitting, solving only elevation, radius, and focal length given the known 10-degree azimuth spacing, initialized from a masked feed-forward pose model (classic SfM is documented to fail on turntable captures: the static background makes it solve for a static camera, and masking removes the features it needs); (3) per-instance surface evidence via a Gaussian-surface backbone with per-image appearance embeddings, used as geometric evidence and initialization, not as the final asset; (4) SUPR-Foot fitting and chart assignment per 4.2 and 4.3; (5) fitting D and appearance per 4.7, image-supervised regime; (6) mesh export with per-instance QA (UDF-style extraction fragility is avoided by the signed field; standard marching cubes suffices), and, later, a feed-forward encoder mapping ring images directly to z so new shoes need no per-instance optimization. The paper shows instantiation from 2, 4, 8, and 36 views in a single figure to demonstrate the method is not StockX-specific; this stays one figure, not a second contribution.

4.10 Design risks and fallbacks

Four, each with a fallback. Chart degeneracy at heels and platforms: three candidate charts compared in Stage 1, heel block reported separately, residual components as the escape hatch if no chart handles it cleanly. SUPR fitting brittleness inside closed shoes: exposed deliberately in Stage 1 week one; if containment-based fitting is unstable, fall back to insole-plane plus landmark initialization with the shape prior tightened. Thin structures: laces are representable but not recoverable from ring views and are excluded from claims; demonstrations use Crocs holes and sandal straps; if thin straps break the signed field, the fallback is a metric-phase style field decoupling distance from sign, adopted only if forced. Factorization leakage (structure and appearance codes entangling): accepted if mild, since no headline claim depends on perfect disentanglement.

5 Contributions and Novelty

Stated as they will appear in the paper, with the novelty defense attached.

1. **A functional 3D representation for topology-varying footwear.** The first body-part-anchored model of a thick, volumetric worn object with a meaningful interior. Novelty defense: garment methods (SMPLicit, DrapeNet, ISP) model thin shells and no cavity; foot-model work never uses the foot as a coordinate system for another object; MEGANE and hand-object work anchor rigid or simple objects without topology variation.
2. **Existence-aware functional correspondence.** Correspondence across objects that do not share topology, defined on shared anatomy, with $\emptyset$ as a legal answer. Novelty defense: partial functional maps handle non-existence driven by shape similarity; ours is driven by functional coverage over shared anatomy, evaluated with a purpose-built annotated benchmark.
3. **FOOTWEAR4K: a category-deep, anatomy-organized footwear dataset.** 4,000+ real instances where the unit of contribution is the registered, canonicalized instance (geometry, appearance, functional coordinates, coverage, structure, partial correspondence), not the raw image; plus a fully scanned physical benchmark set for public release. Novelty defense: ABO

and MVImgNet have scale but no category structure, correspondence, or anatomy; we do not compete on image count and never lead the abstract with “4,000 shoes.”

4. **The ring-observed setting and a blind-zone evaluation protocol.** Formalizing reconstruction under systematic (not accidental) missing coverage, with region-split metrics (visible, top, sole reported separately) and controllable ring elevation in synthetic experiments. Novelty defense: prior sparse-view and amodal work fills random occlusions and lacks any protocol for never-observed regions.

What we deliberately do not claim: relightable PBR as a headline (asset quality, not science; it stays out of the abstract and is attempted only after the core is stable, starting from per-view illumination embeddings); personalized fit prediction (we have no true cavity, last, padding, or pressure data; interior outputs are claimed only as a plausible anatomical envelope, and foot-conditioned retargeting appears as a qualitative application); and any “first ever” absolutes.

6 Experiment Plan

The plan is hypothesis-first. We do not touch the 4,000 StockX instances until the core hypothesis survives a controlled test.

6.1 Stage 1: the make-or-break experiment

Question. Does the anatomical coordinate system actually create a better category space, or does it merely sound good?

Data. The ~20 high-fidelity CAD ground-truth meshes already collected, audited this week for: topology coverage (buy missing categories from the CAD site immediately; this is faster than physical procurement), mesh sanity (non-watertight CAD meshes handled via fast winding numbers rather than repair), scale normalization by insole length, part structure (multi-part models let us bootstrap functional-zone labels from part names), and per-mesh license archiving (CAD marketplace licenses almost certainly forbid redistribution, so this set is internal-only; the public benchmark comes from our own physical scans, whose procurement continues in parallel).

Design. Fit SUPR-Foot into every mesh. Train three models that are identical in architecture, latent size, training data, and optimization, differing only in the input coordinates, in the auto-decoder style of DeepSDF:

- A. Object-centric field: raw coordinates, per-shoe latent.
- A⁺. Aligned object-centric field: same, after best non-anatomical rigid/similarity alignment (ICP to a mean shoe or PCA axes). This row exists to kill the confound that any win comes from alignment rather than anatomy.
- B. Anatomical field: coordinates $\Phi_F(x)$.

Metrics. Four axes. (1) Fit: reconstruction error on all training shoes. (2) Generalization, the decisive one: leave-one-topology-out; train without sandals, test on sandals; the anatomical model should transfer because plantar support and dorsal straps occupy anatomical locations it has already learned, while the object-centric models have no such structure. (3) Interpolation: latent walks from sneaker to sandal, judged on structural plausibility (coverage dissolving into straps versus melted geometry). (4) Correspondence: functional landmark error at shared anatomical queries across shoes.

Decision rule, fixed in advance. If B clearly beats A and A⁺, especially on leave-one-topology-out, the hypothesis holds and we go full speed. If B only ties or wins marginally, the foundation of the story does not exist; we stop before the 4,000 instances become sunk cost and rethink, and we retarget the next deadline rather than force this one.

6.2 Stage 2: correspondence

Freeze the functional-zone annotation schema now, while the set is small: two students independently label functional zones and existence flags (plantar support, toe enclosure, heel enclosure, dorsal coverage, heel retention, ankle enclosure) on the Stage-1 shoes; measure inter-annotator agreement; write the ruling for boundary cases (is a sandal’s heel strap “heel retention: strap” or $\emptyset$?) into a protocol document. Deliverables: region transfer accuracy and existence-prediction F1 for model B versus baselines. Reworking an ambiguous schema at 4,000 scale costs a hundred times more than fixing it at 20.

6.3 Stage 3: ring-to-latent

Two tracks. Synthetic first: render 36-view rings from the CAD meshes with known cameras and lighting; this gives Task C with exact blind-zone ground truth and makes ring elevation a controllable ablation variable (0, 15, 30 degrees), which real StockX data can never provide. Then real: the full pose-and-reconstruction pipeline on a StockX subset, validated against the synthetic findings. Success criterion for the pose solver: silhouette reprojection IoU above roughly 0.9; below that, add per-view refinement or fall back to feed-forward initialization.

6.4 Stage 4: scale and benchmark

Process the 4,000 instances, fine-tune TRELLIS and Amodal3R on the shoe category as baseline rows (budget: about one GPU-day per fine-tune plus preprocessing; start from the most homogeneous subset, watch for the documented overfitting after a few thousand iterations), and assemble the benchmark:

- Task A, category modeling: complete shape to latent to reconstruction across topologies.
- Task B, functional correspondence with $\emptyset$ allowed, scored on the annotated set.
- Task C, ring-to-complete: region-split errors E_{visible} , E_{top} , E_{sole} . This split is the honest headline table: the anatomy prior must show its value precisely where the camera never looked, against the generic model, the shoe fine-tune, and ablations adding category and anatomy priors stepwise.
- Task D, cross-topology generalization: leave-one-topology-out at benchmark scale.
- Task E, foot retargeting: change SUPR-Foot shape parameters, observe style-preserving functional geometry change; qualitative is acceptable.

Physical-scan benchmark: 30–50 purchased shoes with full-coverage capture including sole and top-down, released publicly; optionally CT scans of 5–10 shoes as a memorable interior sanity check (not a fit claim). Licensing of the StockX-derived release: derived assets and pipeline, not raw images; clear with the university before the paper promises release, and state the license explicitly.

References

- [1] A. Osman, T. Bolkart, D. Tzionas, M. Black. SUPR: A Sparse Unified Part-Based Human Representation. ECCV 2022.
- [2] O. Boyne, J. Charles, R. Cipolla. FIND: An Unsupervised Implicit 3D Model of Articulated Human Feet. BMVC 2022.
- [3] O. Boyne et al. FOUND: Foot Optimization with Uncertain Normals for Surface Deformation. WACV 2024.
- [4] O. Boyne, R. Cipolla. FOCUS: Multi-View Foot Reconstruction from Synthetically Trained Dense Correspondences. 3DV 2025.
- [5] E. Corona, A. Pumarola, G. Alenyà, G. Pons-Moll, F. Moreno-Noguer. SMPLicit: Topology-Aware Generative Model for Clothed People. CVPR 2021.
- [6] L. De Luigi, R. Li, B. Guillard, M. Salzmann, P. Fua. DrapeNet: Garment Generation and Self-Supervised Draping. CVPR 2023.
- [7] R. Li, B. Guillard, P. Fua. ISP: Multi-Layered Garment Draping with Implicit Sewing Patterns. NeurIPS 2023.
- [8] J. Li, S. Saito, T. Simon, S. Lombardi, H. Li, J. Saragih. MEGANE: Morphable Eyeglass and Avatar Network. CVPR 2023.
- [9] Z. Fan et al. HOLD: Category-Agnostic 3D Reconstruction of Interacting Hands and Objects from Video. CVPR 2024.
- [10] Y. Ye et al. G-HOP: Generative Hand-Object Prior for Interaction Reconstruction and Grasp Synthesis. CVPR 2024.
- [11] S. Shafique, B. Kong, S. Kong, C. Fowlkes. Creating a Forensic Database of Shoeprints from Online Shoe-Tread Photos (ShoeRinsics). WACV 2023.
- [12] B. Chen et al. ShoeModel: Learning to Wear on the User-Specified Shoes via Diffusion Model. ECCV 2024.
- [13] P. Shrestha et al. 3D Reconstruction of Shoes for Augmented Reality. arXiv:2501.18643, 2025.
- [14] J. Collins et al. ABO: Dataset and Benchmarks for Real-World 3D Object Understanding. CVPR 2022.
- [15] X. Han et al. MVIImgNet2.0: A Larger-Scale Dataset of Multi-View Images. SIGGRAPH Asia / ACM TOG 2024.
- [16] J. Xiang et al. Structured 3D Latents for Scalable and Versatile 3D Generation (TRELLIS). CVPR 2025.
- [17] T. Wu, C. Zheng, F. Guan, A. Vedaldi, T.-J. Cham. Amodal3R: Amodal 3D Reconstruction from Occluded 2D Images. ICCV 2025.
- [18] K. Zhang et al. GS-LRM: Large Reconstruction Model for 3D Gaussian Splatting. ECCV 2024.
- [19] J. Park, P. Florence, J. Straub, R. Newcombe, S. Lovegrove. DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. CVPR 2019.
- [20] E. Rodolà, L. Cosmo, M. Bronstein, A. Torsello, D. Cremers. Partial Functional Correspondence. Computer Graphics Forum 2017.
- [21] B. Huang et al. 2D Gaussian Splatting for Geometrically Accurate Radiance Fields. SIGGRAPH 2024.
- [22] D. Chen et al. PGSR: Planar-Based Gaussian Splatting for Efficient and High-Fidelity Surface Reconstruction. 2024.

- [23] Kim et al. RotGS: Rotation-Guided 3D Gaussian Splatting for Turntable Sequences without Structure-from-Motion. Computer Graphics Forum 2025.
- [24] C. Xu, J. Kerr, A. Kanazawa. Splatfacto-W: A Nerfstudio Implementation of Gaussian Splatting for Unconstrained Photo Collections. arXiv:2407.12306, 2024.
- [25] J. Kulhánek et al. WildGaussians: 3D Gaussian Splatting in the Wild. NeurIPS 2024.
- [26] J. Wang et al. VGGT: Visual Geometry Grounded Transformer. CVPR 2025.
- [27] B. Duisterhof et al. MAST3R-SfM: A Fully-Integrated Solution for Unconstrained Structure-from-Motion. 3DV 2025.
- [28] M. Oechsle, S. Peng, A. Geiger. UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields. ICCV 2021.
- [29] P. Wang et al. NeuS: Learning Neural Implicit Surfaces by Volume Rendering. NeurIPS 2021.
- [30] B. Guillard, F. Stella, P. Fua. MeshUDF: Fast and Differentiable Meshing of Unsigned Distance Field Networks. ECCV 2022.
- [31] F. Hou et al. Robust Zero Level-Set Extraction from Unsigned Distance Fields (DCUDF). ACM TOG 2023.